ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q3, Q4

Question 1

Given: sensor 1280 px wide, pixel pitch p = 5 µm (square), lens f = 16 mm, standoff Z = 400 mm; force sensor 10 mV/N, 12-bit ADC over 0–5 V; plant $\dot{x} = -2x + u$ with zero-order hold, T = 0.1 s.

(a) Assumptions: pinhole (thin-lens) camera model, lens focused on the belt plane, belt plane perpendicular to the optical axis, no lens distortion, so every pixel images the same length on the belt.

(i) The focal length in pixel units is the physical focal length divided by the pixel pitch: $f_x = f/p = (16\times10^{-3}\text{ m})/(5\times10^{-6}\text{ m}) = 3200$. The metres cancel so the answer is in pixels. **f_x = 3200 px**.

(ii) Width of the sensor = N·p = 1280 × 5 µm = 6400 µm = 6.4 mm. Using similar triangles between the sensor and the belt plane, $W_s/f = FOV/Z$, hence $FOV = W_s \cdot Z/f = (6.4\text{ mm})(400\text{ mm})/(16\text{ mm}) = 160$ mm. **FOV = 160 mm**.

Ground sampling distance = FOV/N = 160 mm / 1280 px = 0.125 mm/px. Cross-check with the pixel-based formula Z/f_x = 400 mm / 3200 px = 0.125 mm/px; both agree, so **1 pixel = 0.125 mm = 125 µm** on the belt.

(b) Assumptions: ideal 12-bit ADC (4096 codes), the full 0–5 V range is used, no offset or gain error, and sensor noise is below one LSB so quantisation sets the limit.

Voltage step: 1 LSB = $V_{FS}/2^N = 5\text{ V}/4096 = 1.2207\times10^{-3}$ V = 1.2207 mV.

The sensor maps force to voltage by V = S·F with S = 10 mV/N, so $\Delta F_{min} = \Delta V/S$ = 1.2207 mV / (10 mV/N) = 0.12207 N. **$\Delta F_{min} \approx 0.122$ N**. (Using 4095 steps instead of 4096 codes gives 0.1221 N, the same at 3 s.f.)

(c) With u held constant over [kT, (k+1)T] (zero-order hold) the variation-of-constants formula gives $x((k+1)T) = e^{AT}x(kT) + \int_0^T e^{A(T-s)}B\,u(kT)\,ds$. Since u(kT) is constant it leaves the integral, so $F = e^{AT}$ and $G = \int_0^T e^{A\sigma}B\,d\sigma = A^{-1}(e^{AT} - I)B$ for invertible A.

Substitutting A = −2, B = 1, T = 0.1 s: $F = e^{-0.2} = 0.81873$. $G = (0.81873 - 1)/(-2) = 0.090635$.

Sanity check: the DC gain of the continuous plant is $-A^{-1}B = 0.5$, and the discrete DC gain $G/(1 - F)$ = 0.090635/0.18127 = 0.5 ✓.

$x_{k+1} = 0.8187\,x_k + 0.0906\,u_k$

Question 3

Given: A = [0, 1; 0, −2], B = [0; 1], C = [1, 0], f_s = 1 kHz so the sample time is $T = 1/f_s$ = 1 ms.

(a) Reason 1 – noise amplification: the backward difference $\hat{\omega}_k = (\theta_k - \theta_{k-1})/T$ has frequency response $(1 - e^{-j\omega T})/T$, whose magnitude grows with frequency (≈ ω for small ωT), i.e. it is a high-pass operation. Any noise on the encoder signal therefore appears in the velocity multiplied by roughly 1/T = 1000 s^{-1}, and can swamp the true velocity.

Reason 2 – quantisation: the encoder only reports integer counts, so Δθ is an integer multiple of the count size q. Dividing by T = 1 ms gives velocity steps of q/T; for a 4096-count encoder $q = 2\pi/4096 = 1.53\times10^{-3}$ rad, so one step is 1.53 rad/s. At low speed the estimat flips between 0 and one step (a noisy staircase), exactly where the damping term of the PD law is needed.

(A further, smaller issue is the half-sample delay of a backward difference, but the two reasons above are the main ones.)

(b) The analogue prototype is the first-order low-pass $H(s) = \omega_c/(s + \omega_c)$ with $\omega_c = 2\pi f_c = 2\pi \times 20 = 125.66$ rad/s. The bilinear (Tustin) transform replaces s by $(2/T)(z - 1)/(z + 1)$ with $T = 1/f_s = 0.001$ s.

Define $a = \omega_c T/2 = 125.66 \times 0.001/2 = 0.062832$. Then $H(z) = \omega_c / [(2/T)(z-1)/(z+1) + \omega_c] = a(z + 1) / [(1 + a) z - (1 - a)] = a(1 + z^{-1}) / [(1 + a) - (1 - a) z^{-1}]$.

Dividing through by $(1 + a)$: $a/(1 + a) = 0.062832/1.062832 = 0.059117$ and $(1 - a)/(1 + a) = 0.937168/1.062832 = 0.881767$. Hence with u = differenced signal and y = filtered velocity: $\mathbf{y_k = 0.05912\ (u_k + u_{k-1}) + 0.88177\ y_{k-1}}$. (Frequency pre-warping is neglected since $f_c \ll f_s/2$.)

Verification of the DC gain: set u and y constant ($z = 1$): $y = 2(0.05912)u + 0.88177\ y \Rightarrow y(1 - 0.88177) = 0.11824\ u \Rightarrow y/u = 0.11824/0.11823 = 1.000$. In general $H(1) = 2a/[(1 + a) - (1 - a)] = 2a/2a = 1$ exactly, so the steady-state output equals the commanded input ✓.

(c) $A - LC = [-\ell_1, 1; -\ell_2, -2]$; characteristic polynomial $\det(sI - A + LC) = (s + \ell_1)(s + 2) + \ell_2 = s^2 + (2 + \ell_1)s + (2\ell_1 + \ell_2)$.

Desired $(s + 10)^2 = s^2 + 20s + 100$. Then $2 + \ell_1 = 20 \Rightarrow \ell_1 = 18$, and $2\ell_1 + \ell_2 = 100 \Rightarrow \ell_2 = 100 - 18 = 82$.

$\mathbf{L = [18, 82]^T}$.

(d) Advantage of the observer that no filter of this kind can give: the observer is driven by both the measurement y and the model with the actual input u, so it predicts the motion between samples. The velocity estimate therefore has little or no phase lag. The low-pass filter on differenced data, by contrast, always pays a lag of about $1/(2\pi f_c) = 8$ ms for its noise rejection, a trade-off it cannot escape. In addition the observer gives all states, and can be augmented with a disturbance state (e.g. unknown load torque) which no fliter on θ can estimate.

Circumstance where the filter is safer: when the plant model is poorly known or changes during operation, for example an unknown payload, varying friction, contact with the environment, or actuator saturation making u differ from the commanded value. The observer propagates the wrong model so its estimate can be biased (and in the worst case unstable with the feedback loop), whereas the model-free fliter just smooths what the encoder actually measures.

Question 4

Given: plant $\theta\ddot{} + a\theta\dot{} + c\theta = b\,u$ with a = 2, c = 6, b = 1; controller $u = N_{ff}\,r - k_1\theta - k_2\theta\dot{}$.

(a) Substitute the control law into the plant: $\theta\ddot{} + a\theta\dot{} + c\theta = b\,N_{ff}\,r - b\,k_1\,\theta - b\,k_2\,\theta\dot{}$. Moving the feedback terms to the left: $\theta\ddot{} + (a + b\,k_2)\,\theta\dot{} + (c + b\,k_1)\,\theta = b\,N_{ff}\,r$.

Numerically, $a + b\,k_2 = 2 + 2 = 4$ and $c + b\,k_1 = 6 + 2 = 8$, so the characteristic equation is $s^2 + 4s + 8 = 0$. Roots: $s = [-4 \pm \sqrt{(16 - 32)}]/2 = -2 \pm 2j$. They have negative real parts, so the loop is stable; $\omega_n = \sqrt{8} = 2.83$ rad/s and $\zeta = 4/(2 \times 2.83) = 0.707$.

Transfer function from r to θ: $T(s) = b\,N_{ff} / (s^2 + 4s + 8)$. DC gain ($s = 0$): $T(0) = N_{ff}/8$. For unity DC gain, $\mathbf{N_{ff} = 8}$. (Note that this is larger than 1 because the gravity-related stiffness c and the feedback k_1 both resist the displacement.)

(b) With the actual b = 0.75 and unchanged $N_{ff} = 8$, $k_1 = k_2 = 2$, the closed loop becomes $\theta\ddot{} + (a + bk_2)\theta\dot{} + (c + bk_1)\theta = bN_{ff}\,r$, i.e. $\theta\ddot{} + 3.5\theta\dot{} + 7.5\theta = 6\,r$.

For a unit step the steady state is found by setting derivatives to zero: $7.5\,\theta_{ss} = 6 \times 1$, so $\theta_{ss} = 6/7.5 = \mathbf{0.8}$.

Tracking error $= (r - \theta_{ss})/r \times 100\% = (1 - 0.8) \times 100 = \mathbf{20\%}$. A common mistake would be to say the output falls by the same 25% as the actuator gain; that is wrong here, because b also appears in the feedback stiffness $c + bk_1$, which decreases from 8 to 7.5 and partially compensates. The poles also change: $s^2 + 3.5s + 7.5 \rightarrow s = -1.75 \pm 2.11j$, still stable.

(c) Take the state $z = [\theta, \dot{\theta}, \xi]^T$. The plant gives $\ddot{\theta} = -a\dot{\theta} - c\theta + b\,u$ with $u = -k_1\theta - k_2\dot{\theta} + K_I\xi$, and $\dot{\xi} = r - \theta$. With $r = 0$ (r only shifts the equilibrium) the closed-loop matrix is $A_{cl} = [0, 1, 0; -(c + bk_1), -(a + bk_2), bK_I; -1, 0, 0]$.

Characteristic polynomial $\det(sI - A_{cl})$, expanding along the first row: $s\cdot[(s + a + bk_2)s - 0] + 1\cdot[(c + bk_1)s + bK_I] = s^3 + (a + bk_2)s^2 + (c + bk_1)s + bK_I$. **(Result)**

Desired poles -4 and $-2 \pm 2j$: $(s + 2 - 2j)(s + 2 + 2j) = (s + 2)^2 + 4 = s^2 + 4s + 8$. Multiply by $(s + 4)$: $s^3 + 4s^2 + 8s + 4s^2 + 16s + 32 = s^3 + 8s^2 + 24s + 32$.

Compare coefficients with $a = 2$, $b = 1$, $c = 6$: s^2: $2 + k_2 = 8 \Rightarrow \mathbf{k_2 = 6}$. s^1: $6 + k_1 = 24 \Rightarrow \mathbf{k_1 = 18}$. s^0: $b\,K_I = 32 \Rightarrow \mathbf{K_I = 32}$.

Check: all coefficients of the desired polynomial are positive and $(8)(24) = 192 > 32$, so Routh–Hurwitz confirms the three poles are in the left half plane ✓.

(d) Because the closed loop is stable, the steady-state error is zero for any b and c.

Failure mode: integrater windup. Remedy: clamp the integrator output (anti-windup).

Note: the integrator adds one state, so the loop becomes third order.